

Document Detail

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Title: AmiCORE Software Version 2.1.85.752 (2.1M) Installation

Owner: 180305 Jeffery Mashke

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Reference

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DRW-6112831940 [A]	DOCUMENT	Related	No	CURRENT
Title:USB WITH SOFTWARE AmiCORE 2.1M				
SDL-SDL12830001 [AZ]	DOCUMENT	Related	No	CURRENT
Title:AmiCORE Service Document Listing (SDL)				
888-REP-089654 [A]	DOCUMENT	Related	No	CURRENT
Title:AmiCORE SW 2.1M Systems Verification Report				
PKG-006302 [A]	DOCUMENT	Related	No	INWORKS
Title:AmiCORE SW 2.1M (KabiTrack # 1849528)				

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Note To Approver:

Note From Approver:Approved

MODEL #: 6R8800**SUBJECT:** AmiCORE Software Version 2.1.85.752 (2.1M) Installation

1.0 PURPOSE

The purpose of this Technical Service Bulletin (TSB) is to provide information on the installation of AmiCORE Software Version 2.1.85.752. This software may also be referred to as 2.1M.

2.0 SCOPE

This technical service bulletin provides information, part numbers, and instructions for the installation of AmiCORE Software Version 2.1M for devices within Korea currently at Software Version 2.1 or later (up to and including 2.1.82.735 (2.1L)). Software Version 2.1.85.752 may be abbreviated to 2.1M in some documentation.

Software version 2.1M is released for implementation solely in Korea.

3.0 BACKGROUND

Software version 2.1M is an updated version of the previous software 2.1L with the following changes:

- Disinfection Timer option removed from Collection Parameters screen 5.
The removal of this parameter is the result of regional requirements within Korea.
- Addition of Korean language to the UI
- Minor anomaly fixes
- Optional Service feature for displaying additional events related to kit installation checks

The Software Version 2.1M contains the following alerts above and beyond those listed in the AmiCORE Service Manual:

Code	Description	Resolution
1003	Incorrect Date/Time The date/time is set incorrectly or a hardware issue exists.	Adjust the date/time. If this issue persists, call a qualified service representative. If running a procedure, log file data may not be saved if the date/time is not corrected.
1103	Admin Parameter Error The system detected an error with Admin parameters.	Power OFF the instrument. Call a qualified service representative.

Code	Description	Resolution
2031	Leak Detected The system is unable to build pressure within the kit. The cassette will automatically unload.	Ensure the green clamp on the PAS line is closed. Check for leaks on the Plasma and PAS Y junction. If a leak is found, press the <i>STOP button</i> and end the procedure. Unload the kit and start over with a new kit.
2032	Weight on Scale The system detected a large weigh scale change, possibly caused by the addition of the PAS container.	Remove the PAS container from the weigh scale. Wait until system prompts you to install and hang the PAS container. Touch the <i>check button</i> to continue.
3013	Scale Instability Detected The system detected unusual weigh scale instability.	Ensure the containers are hanging freely on the red and yellow hooks and that no air is blowing toward the device.
3106	Centrifuge Line Blockage The system detected a possible PRP centrifuge line occlusion.	Ensure there are no kinks or air in the PRP line coming from the separation chamber. Touch the <i>check button</i> to continue. If the issue persists, press the <i>STOP button</i> and end the procedure.
3509	No Blood Flow (Return Prime) Vein pressure reached the minimum allowed pressure.	Check Pressure Sensors and O-rings if issue persists.
4006	Clamp Error The system did not detect fluid flow from the PAS container	Ensure the green clamp on the PAS line is open. Ensure the cannula on the PAS container is properly broken. Ensure there are no kinks on the PAS line. Touch the <i>check button</i> to continue.
4008	Clamp Error The system detected fluid flow from the plasma container instead of the PAS container.	Ensure the yellow clamp on the plasma line is closed and the green clamp on the PAS line is open. Touch the <i>check button</i> to continue.
4009	Product Transfer Error The system detected a PAS flow issue while transferring the product to the platelet storage bag.	Ensure there are no kinks on the fluid pathway from the PAS container to the platelet storage container. Ensure the green clamp on the PAS line is completely open. Touch the <i>check button</i> to continue.
4010	Low PAS Volume The system detected that the PAS container volume is low.	Monitor the PAS container and be prepared to install a new PAS container if needed. Touch the <i>check button</i> to continue.

Code	Description	Resolution
4011	Storage Volume Too Low The volume of plasma in the storage fluid pathway exceeds the minimum the system can deliver. PAS ratio will be maintained, as a result, storage fluid volume will be higher than targeted.	Touch the <i>check button</i> to continue.
10016	Centrifuge Speed Error Centrifuge door commanded to unlock, but centrifuge commanded to spin. Centrifuge door will proceed to unlock when the centrifuge stops spinning.	Wait for centrifuge to spin down and the system can be resumed.
10017	Scale Instability Error The system detected unusual instability on the weigh scales during the previous procedure.	Ensure that there is sufficient space for containers to be able to hang freely on the red and yellow hooks and that no air is blowing toward the device. Call a qualified service representative if the issue persists.
10205	Centrifuge Door Lock Error The system detected a fault with the centrifuge door lock.	Check centrifuge door to ensure working properly.

Part Information:

Spare Part	Part Number	Notes
USB with Software AmiCORE 2.1M	6112831940	USB drive containing upgrade software. May be used for upgrading multiple devices.

4.0 PROCEDURE

Software 2.1M Installation:

- 4.1 Power the AmiCORE On.
- 4.2 Optional: export all data logs and procedure records for sharing with R&D.
- 4.3 Carefully insert the version 2.1M USB drive in rear USB port (either port).
- 4.4 Select the I button.
- 4.5 Select the Update button.
- 4.6 Enter the service password "PASSWORD2".
- 4.7 The software upgrade will begin. When the I button screen returns, press the stop button and power the AmiCORE off.
- 4.8 Power the AmiCORE on.
- 4.9 The software update will continue. When the screen instructs "Please power cycle the system", power the AmiCORE off.
- 4.10 Power the AmiCORE on.
- 4.11 Perform power up stop button check.
- 4.12 Perform power up audio check.
- 4.13 Enter service mode.
- 4.14 Select the I button.
- 4.15 Select the System Info button.
- 4.16 Select the Instrument Configuration tab.
- 4.17 Verify the MPU SW Version is 2.1.85.752.
- 4.18 Select the check button.
- 4.19 Select the check button to return to the service menu.
- 4.20 Select the Centrifuge tab.



4.21 Select the Collection Parameters button.

Verify the default values for the Collection Parameters: There are thirteen screens of parameters. All parameters must be set to default value listed below unless noted within description or per direction of GTM

Screen	Name	Default	Units	Description
1	Inlet Hematocrit	35	%	Target centrifuge hematocrit. Do not adjust without the authorization of R&D.
1	Default Setpoint	52		Target interface setpoint. May be adjusted to address product issues. Recommend defaulting at 56 for newly placed instruments.
1	Excess Plasma Return Volume	50	mL	Inter-cycle plasma overcollection limit. Lower values will cause more interruptions to a procedure.
1	IFC Rate Timer	120	s	Time between rate change. May be adjusted per customer requirements. See section 5.0.
1	Enable DXT	Disabled		Enable/Disable use of DXT. Setting carried over from previous release during upgrade.

Screen	Name	Default	Units	Description
1	Undo Draw Increase	15	s	Window of time after IFC automatically increases the draw rate where, upon detection of an occlusion, the system will return back to the previous draw rate instead of dropping below it. May be adjusted per customer requirements. See section 5.0
1	PAS Prime Rate	70	mL/min	PAS prime rate. Do not adjust without the authorization of R&D.
1	PAS Prime Threshold	4	g	Minimum weight change to indicate bag on scale 1. Do not adjust without the authorization of R&D.
1	IFC Decrease Count	3		Number of IFC events before drop target draw rate 10 mL/min. May be adjusted per customer requirements. See section 5.0
1	IFC Pause	6.0	s	Pause time for IFC. May be adjusted per customer requirements. See section 5.0.
1	LED Minimum Brightness	280		Lower brightness limit on LED. Do not adjust without the authorization of R&D.
1	Raise Draw Rate	Unchecked	n/a	Setting that raises the draw rate at the end of a draw cycle. May be adjusted per customer requirements. See section 5.0
2	Max Yield	12	x10 ¹¹	Maximum platelet yield. May be adjusted per customer and local regulatory requirements.
2	Max Plasma Product	890	mL	Maximum concurrent plasma. May be adjusted per customer and local regulatory requirements.
2	Max PAS Ratio	99	%	Maximum PAS Ratio. May be adjusted per customer and local regulatory requirements.
2	Max WBCF Rate	70	mL/min	Max processing rate. May be adjusted per customer requirements.
2	Max Cycle Volume	300	mL	Max cycle volume. May be adjusted per customer requirements.

Screen	Name	Default	Units	Description
2	Plasma Xfer Cycles	99.0		Number of remaining cycles in the procedure to start performing plasma overage transfer. Plasma overage transfer helps reduce the volume of plasma necessary to return during reinfusion but increases the frequency of valve and pump switching during the procedure. May be adjusted per customer requirements but recommend a value no lower than 2.5 in order to minimize time impact on reinfusion.
2	Storage Volume Limit	1600	mL	Estimator limit for storage volume. May be adjusted per customer requirements.
2	Normal Setpoint Ramp Rate	20	/100k ticks	The rate for ramping the commanded setpoint to the target setpoint during normal collection. Do not adjust without the authorization of R&D.
2	Spillover Setpoint Ramp Rate	0	/100k ticks	The rate for ramping the commanded setpoint to the target setpoint during a spillover. Do not adjust without the authorization of R&D.
2	Control Rate Calc: Dynamic	Unchecked	n/a	Setting for turning on multiplying the control signal by the commanded centrifuge and recirculation pump rates. Do not adjust without the authorization of R&D.
2	Flow Rate Band	15	mL/min	Flow rate margin used for determining whether the system should recalculate the target setpoint. Do not adjust without the authorization of R&D.
2	Consecutive Air Count	2	n/a	Minimum number of consecutive ticks of air expected during AC prime. Do not adjust without the authorization of R&D.
3	Prime Time	7.5	min	Estimated prime time. Used to display estimated time remaining while priming the disposable kit.
3	Percent Load Fluids	5	%	Indicates the step prime has completed when this percentage is displayed.
3	Percent AC Prime	15	%	Indicates the step prime has completed when this percentage is displayed.
3	Percent Cent Cal	32	%	Indicates the step prime has completed when this percentage is displayed.

Screen	Name	Default	Units	Description
3	Percent Blood Cal	46	%	Indicates the step prime has completed when this percentage is displayed.
3	Percent Saline 1	53	%	Indicates the step prime has completed when this percentage is displayed.
3	Percent Saline 2	64	%	Indicates the step prime has completed when this percentage is displayed.
3	Percent Saline 3	68	%	Indicates the step prime has completed when this percentage is displayed.
3	Percent Saline 4	77	%	Indicates the step prime has completed when this percentage is displayed.
3	Percent Spun Up	95	%	Indicates the step prime has completed when this percentage is displayed.
4	Pressure Threshold	-290	mmHg	Flow control during prime. Do not adjust without the authorization of R&D.
4	Low Flow Rate	60	mL/min	Flow control during prime. Do not adjust without the authorization of R&D.
4	Rate Increment	10	mL/min	Flow control during prime. Do not adjust without the authorization of R&D.
4	Low Rate Pause	20		Flow control during prime. Do not adjust without the authorization of R&D.
4	Increment Pressure	-50	mmHg	Flow control during prime. Do not adjust without the authorization of R&D.
4	Cal Start Volume	5	mL	Pump Calibration factor. Do not adjust without the authorization of R&D.
4	STDev Limit	0.1		Pump Calibration factor. Do not adjust without the authorization of R&D.
4	DynCal Samples	50		Pump Calibration factor. Do not adjust without the authorization of R&D.
4	Unstable Counts	25		Pump Calibration factor. Do not adjust without the authorization of R&D.
4	Samples For CF	10		Pump Calibration factor
4	Samples For STDev	30		Pump Calibration factor
4	Pause Time	30		Pump Calibration factor. Do not adjust without the authorization of R&D.
5	All	Checked		Sets all screen five parameters to checked or unchecked. Note that checking or unchecking this will also set/reset Max Plasma Product and Max PAS Ratio.
5	Arm Entry	Checked		Selection of Arm. May be disabled per customer requirements.

Screen	Name	Default	Units	Description
5	NINO Time	Checked		Needle In Needle Out Time. May be disabled per customer requirements.
5	Status Ind.	Checked		Status Indicator enable. May be disabled per customer requirements.
5	Donor Ind.	Checked		Donor Indicator enable. May be disabled per customer requirements.
5	Divert Vol.	Checked		Divert initial draw prime. May be disabled per customer requirements.
5	PAS Kits	Checked		Enable PAS kits. May be disabled per customer requirements.
5	AC Prime	Checked		AC addition during prime. May be disabled per customer requirements.
5	Update UI	Checked		If enabled, adds Total Plasma Loss calculation. May be disabled per customer requirements.
6	Pls. PC Limit	1		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	Pls. PRP Limit	1		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	WS1 Min Delta	1.500	g	Bag empty detection factor. Do not adjust without the authorization of R&D.
6	IP Min Correct	-1.000		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	IP Max Correct	1.000		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	WS1 Blackout Time	5		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	RBC Min Correct	-1.000		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	RBC Max Correct	1.000		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	IP Vol Normal	50		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	IP Vol High	2000		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	RBC Emp Level	1		Bag empty detection factor. Do not adjust without the authorization of R&D.
6	Reset Press	Checked		Bag empty detection factor. Do not adjust without the authorization of R&D.
7	Blood Pump Ramp Delta	7	mL/min/s	Blood pump ramp rate. May be adjusted per customer requirements. See section 5.0.

Screen	Name	Default	Units	Description
7	Recovery Time	5.0	s	Recovery time for IFC. If second occlusion occurs within Recovery Time, an alert will occur. May be adjusted per customer requirements. See section 5.0
7	Detect Exit to RF	0	%	% remaining in final draw cycle which will initiate reinfusion during reprime or centrifuge line occlusion. May be adjusted per customer requirements to up to 50%.
7	AC Overweight Check Amount	0	g	Used to detect improper sized bag used for AC. Default setting 0g disables feature. For customers wishing to enable feature, see section 6.0.
7	Drop Draw Rate	Unchecked		If checked, IFC will lower draw rate if flow issue occurs within Recovery Time. May be adjusted per customer requirements see Section 5.0.
7	Reprime Slope Threshold	15		Bag empty detection factor. Do not adjust without the authorization of R&D.
7	IFC Pump Rate	75	mL/min	Pump rate to vent during IFC. Do not adjust without the authorization of R&D.
7	IFC End Vent Press	-80	mmHg	Limit to stop IFC vent. Do not adjust without the authorization of R&D.
7	IFC Vent Time	3.0	s	Time to vent during IFC. Do not adjust without the authorization of R&D.
7	PFR CIR Factor	1.80		Prepare for reinfusion CIR. May be set up to 30.00 but should not be decreased without direction from R&D. Higher numbers correspond to faster pump rates at the start of reinfusion (increasing the short-term citrate infusion rate to the donor but potentially decreasing the total reinfusion time). The total CIR averaged over the entire reinfusion process is unchanged by modifying this parameter.
7	PRP Recirc	Disabled		Do not enable without authorization of R&D.
7	RBC Empty Threshold	12		Bag empty detection factor. Do not adjust without the authorization of R&D.
8	SV Transfer Ramp Down Vol	2	mL	Product Transfer storage volume factor. Do not adjust without the authorization of R&D.

Screen	Name	Default	Units	Description
8	Light Adjust Processed Volume	200	mL	WB Processed Volume at which interface brightness adjustment occurs. Do not adjust without the authorization of R&D.
8	Light Adjust Brightness Value	50000		Target LED Brightness during interface brightness adjustment. Do not adjust without the authorization of R&D.
8	Light Adjust Min Brightness	35000		Minimum brightness reading for light adjustment to occur. Do not set above value for Light Adjust Brightness Value. This value may be set from 0-35000 to control the frequency of light adjustments. Do not adjust above 35000 without the authorization of R&D.
8	Light Adjust Brightness Vol	50	mL	Minimum continuous processing volume before a brightness adjustment can occur. Do not adjust without the authorization of R&D.
8	Target Peak To Peak	55706		Feedback brightness reading during Interface Calibration. Do not adjust without the authorization of R&D.
8	PPP Transfer Overpump	0	mL	Excess volume to pump out of collection pack during product transfer. May be adjusted per customer requirements to ensure complete emptying of collection pack.
8	Residual Transfer Vol	10	mL	Volume to pump out of the centrifuge pack when the transfer residual button is pressed. May be adjusted per customer requirements.
8	Recirc Average Method	Amicus		Method for executing recirculation calculation, OLD = AmiCORE SW 2.1E and earlier, AMICUS = 20 second moving average, NEW = 20 second moving average with reset on processing rate changes. Do not adjust without the authorization of R&D.
8	NPC Volume	15	mL	Volume the system attempts to return after each Non Plasma Collect Return to clear the line of lower hematocrit blood. Do not adjust without the authorization of R&D.

Screen	Name	Default	Units	Description
8	Long Spillover Volume	20	mL	Extra volume to recirculate during early procedure spillover corrections . Do not adjust without the authorization of R&D.
8	WB Processed Long Spillover	200	mL	Prior to this threshold, extra volume is pumped to clear redness during a spillover. Do not adjust without the authorization of R&D.
9	Cassette Load Rate	1400	Hz	Cassette Movement Factor. Do not adjust without the authorization of R&D.
9	Cassette Unload Rate	2200	Hz	Cassette Movement Factor. Do not adjust without the authorization of R&D.
9	Cassette Load Pump Rate 1	35	mL/min	Cassette Movement Factor. Do not adjust without the authorization of R&D.
9	Cassette Load Time 1	57	1/10s of a sec	Cassette Movement Factor. Do not adjust without the authorization of R&D.
9	Cassette Load Pump Rate 2	35	mL/min	Cassette Movement Factor. Do not adjust without the authorization of R&D.
9	Cassette Load Time 2	30	1/10s of a sec	Cassette Movement Factor. Do not adjust without the authorization of R&D.
9	Cassette Load Pump Rate 3	35	mL/min	Cassette Movement Factor. Do not adjust without the authorization of R&D.
9	Cassette Load Time 3	50	1/10s of a sec	Cassette Movement Factor. Do not adjust without the authorization of R&D.
9	Cassette Unload Pump Rate	25	mL/min	Cassette Movement Factor. Do not adjust without the authorization of R&D.
9	Cassette Unload Time	100	1/10s of a sec	Cassette Movement Factor. Do not adjust without the authorization of R&D.
10	Free Flow Delta 1	0.3	g	Parameter for empty RBC bag detection during Reinfusion. Do not adjust without the authorization of R&D.
10	Free Flow Delta 2	0	g	Parameter for empty RBC bag detection during Reinfusion. Do not adjust without the authorization of R&D.
10	Free Flow Slope	-5.0	g/min	Parameter for empty RBC bag detection during Reinfusion. Do not adjust without the authorization of R&D.
10	AC Vacuum Delta	140	mmHg	Draw line vacuum to pull before filling with saline. Do not adjust without the authorization of R&D.
10	AC Vacuum Rate	150	mL/min	Pump rate for pulling draw line vacuum in prime. Do not adjust without the authorization of R&D.

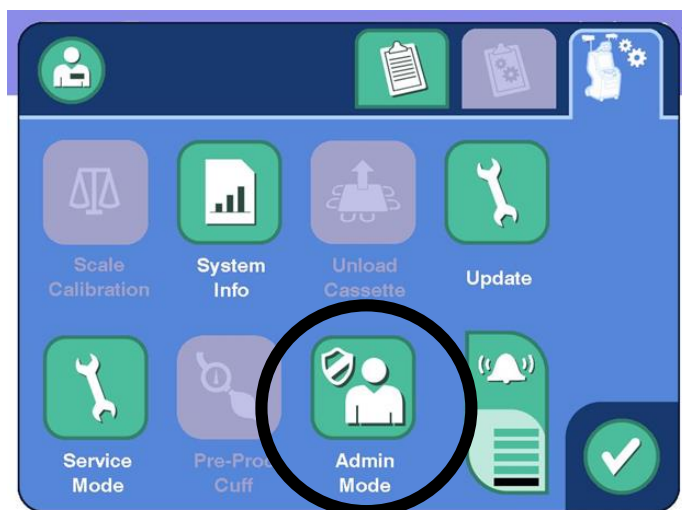
Screen	Name	Default	Units	Description
10	AC Vacuum Stabilize Ticks	40	1/10s of a sec	Wait time for saline to fill the draw line after pulling vacuum. Do not adjust without the authorization of R&D.
10	AC Vacuum Count	3		Number of times to perform draw line vacuum cycle. Do not adjust without the authorization of R&D.
10	Additional SP1_2 Blood Pump Vol	20	mL	Extra volume to pump during a specific prime state. Do not adjust without the authorization of R&D.
10	Additional SP1_2 Cent Pump Vol	0	mL	Extra volume to pump during a specific prime state. Do not adjust without the authorization of R&D.
10	Slow Centrifuge Speed	500	rpm	Centrifuge speed upon entry to Prepare for Venipuncture. Do not adjust without the authorization of R&D.
10	VT Centrifuge Speed	500	rpm	Centrifuge speed to command during Volume Tracking states in prime. Do not adjust without the authorization of R&D.
10	Auto Stop Centrifuge	Checked	n/a	Setting to automatically stop the centrifuge after completion of fluid prime, if the operator hasn't yet advanced to the Venipuncture Screen. May be adjusted per customer requirements.
11	WS1 Low Check Buffer	0	g	Buffer to add to the 2029 alert check during kit installation checks. Do not adjust without the authorization of R&D.
11	WS1 High Check Buffer	40	g	Buffer to add to the 2030 alert check during kit installation checks. Defaulted to 40 to avoid nuisance alerts. Do not adjust without the authorization of R&D.
11	WS2 Low Check Buffer	0	g	Buffer to add to the 2027 alert check during kit installation checks. Do not adjust without the authorization of R&D.
11	Plasma Bag Max Vol	890	mL	Maximum volume that can be collected in the plasma container based on physical size and residual air. Do not adjust without the authorization of R&D.
11	Saline Flush Volume	50	mL	Volume of saline used during reinfusion to rinse the disposable set lines. Do not adjust without the authorization of R&D.

Screen	Name	Default	Units	Description
11	Saline Flush Slow Rate	60	mL/min	Slow pump rate during saline flush states in reinfusion. Do not adjust without the authorization of R&D.
11	Saline Flush Fast Rate	150	mL/min	Fast pump rate during saline flush states in reinfusion. Do not adjust without the authorization of R&D.
11	RTD Multiplier	1.000	n/a	Multiplier to increase final reinfusion speed. Do not adjust without the authorization of R&D.
11	AC Return Prime Rate	40	mL/min	Rate at which to prime the return line with ACD. Do not adjust without the authorization of R&D.
11	PLS Spin Down Vol	5	mL	Estimate of the volume of plasma pushed out of the collection pack when the centrifuge spins down after reinfusion. Do not adjust without the authorization of R&D.
11	Pack Unload Volume	25	mL	Volume to pump out of the separation pack to make it easier to remove the spool. May be adjusted per customer requirements, see section 7.0.
11	Pack Removal Rate	120	mL/min	Flow rate for Pack Unload volume. May be adjusted per customer requirements, see section 7.0.
12	Volume Prior to Plasma Collect	110	mL	Minimum volume of plasma pumped through the collection pack to displace saline from prime with plasma. Do not adjust without the authorization of R&D.
12	RBC Bag Vol Min	35	mL	Minimum buffer volume in the RBC container at the end of a return cycle to avoid emptying of the RBC bag. Do not adjust without the authorization of R&D.
12	IP Bag Vol Min	30	mL	Minimum buffer volume in the IP container at the end of a return cycle to avoid emptying of the IP bag. Do not adjust without the authorization of R&D.
12	Blood Pump Cal Hct	WB Hct Avg		Hematocrit estimate to use in blood pump calibration adjustment equation. Do not adjust without the authorization of R&D.

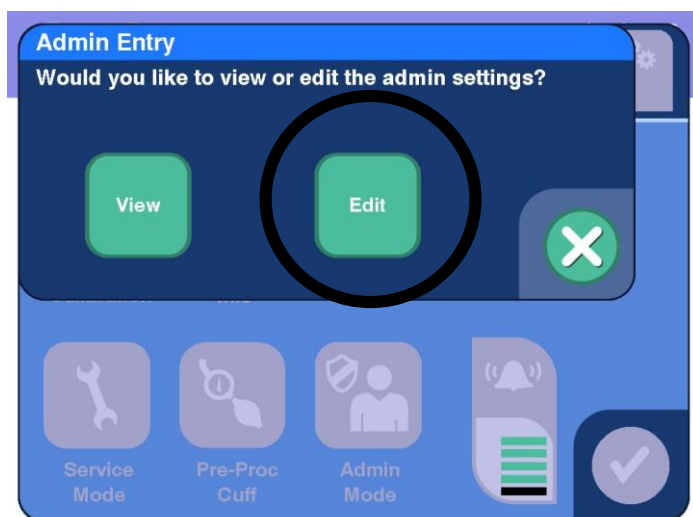
Screen	Name	Default	Units	Description
12	Blood Pump CF Adjuster	0		Adjustment to the blood pump correction factor. Do not adjust without the authorization of R&D.
12	Blood Pump Cal AC Factor	1		Adjustment to the estimated AC volume pumped by the blood pump during a given cycle. Do not adjust without the authorization of R&D.
12	Min Plasma Vol After Overage Xfer	20	mL	Desired minimum volume in the plasma bag at end of overage transfer, intended to prevent emptying of plasma bag at the beginning of a procedure. Do not adjust without the authorization of R&D.
12	Min Plasma % After Overage Xfer	100	%	Desired minimum % of total plasma collection volume in the plasma bag at end of overage transfer, raise to start overage transfer later in the procedure. Do not adjust without the authorization of R&D.
12	Min Plasma Xfer Vol	10	mL	Minimum volume of plasma transferred during Non-Plasma Collect and Plasma Overage Transfer states. Do not adjust without the authorization of R&D.
12	Plasma Xfer Vol Plus – Return	5	mL	Additional minimum volume transferred during plasma overage transfer when triggered at intervals during return cycles. Do not adjust without the authorization of R&D.
12	Plasma Xfer Vol Plus – Draw	3	mL	Additional minimum volume transferred during plasma overage transfer when triggered at intervals during draw cycles. Do not adjust without the authorization of R&D.
12	PPP Undercollect Margin	3	mL	Allowable undershoot of plasma volume during plasma overage transfer. Do not adjust without the authorization of R&D.
13	Max PPP Over-Collect Margin	15	mL	Maximum volume of extra plasma in a given return cycle that can be transferred to the RBC bag during subsequent draw/return cycles. Do not adjust without the authorization of R&D.

Screen	Name	Default	Units	Description
13	Blood Pump Cal CF Contribution	0.50		Fraction of contribution from WB Hct Avg vs. Donor Hct in the blood pump correction factor recalculation. Do not adjust without the authorization of R&D.
13	Debug Kit Install Checks	Unchecked		Display of passing kit installation check data. Do not adjust without the authorization of R&D.

- 4.22 Select the check button.
- 4.23 Power the AmiCORE Off.
- 4.24 Power the AmiCORE On.
- 4.25 Perform stop button test when prompted.
- 4.26 Acknowledge audio test when prompted.
- 4.27 Select the I button.
- 4.28 Select the Admin Mode button.



- 4.29 Select the Edit option.



- 4.30 Enter the Admin Password: "ADMINPASSWORD".
(Note: the password is case sensitive.)
- 4.31 Select the second tab (icon of a barcode).
- 4.32 Select the down arrow five times to display screen 6/6.
- 4.33 Select the fourth tab (icon of a bag and gears).
- 4.34 Select the down arrow three times to display screen 4/4.
- 4.35 Select the fifth tab (icon of clipboard and gears).
- 4.36 Select the down arrow four times to display screen 5/5.
- 4.37 Select the check button.
- 4.38 Power the AmiCORE off.
- 4.39 Remove the USB drive.
- 4.40 Perform software upgrade tests per AmiCORE Field Repair Test Matrix.

5.0 INTELLIGENT FLOW CONTROL (IFC) PARAMETERS

IFC Rate Timer, IFC Decrease Count, and IFC Pause are service settings that may be adjusted to meet customer requirements with regards to Intelligent Flow Control (IFC). The default values are set to limit the occurrence rate of occlusion (3508/3509) alerts while balancing against maximizing the donor draw rate and minimizing procedure time.

For those customers who experience a low rate of 3508/3509 alerts and wish to further maximize donor draw rate and minimize procedure time, the IFC Decrease Count may be raised, the IFC Pause may be lowered, the IFC Rate Timer may be lowered, Blood Pump Ramp Delta may be raised, and Raise Draw Rate may be enabled as recommended below.

Undo Draw Increase may be set to 0 to never undo a draw rate increase after good flow, even if a IFC is triggered, or may be increased to 90 to undo a draw rate increase if an IFC is triggered. If the increased draw rate (5 mL/min) is simply reversed within the Undo Draw Increase window, that may result in an overall faster rate compared to the case where the Undo Draw Increase is set to 0 and IFC reduces the draw rate by 10 mL/min.

Name	Min – Max	Default Setting	Recommended Range	Recommended Maximized-Draw Rate Setting
IFC Decrease Count	0 – 10	3	2 – 6	6
IFC Pause	0.0 – 20.0	6.0	4.0 – 7.0	3.0
IFC Rate Timer	0.0 – 1000.0	120.0	60.0 – 120.0	60.0
Blood Pump Ramp Delta	1 – 10	7	3 – 10	10.0
Raise Draw Rate	Enabled / Disabled	Disabled (unchecked)	n/a	Enabled (checked)
Undo Draw Increase	0 – 255	15 – recommend setting to 60	0 – 90	90

With the above recommended settings for maximizing the draw rate, customers may experience, on average, slightly higher donor rates and slightly shorter procedure times, but operators may have to respond to more 3508/3509 alerts. The recommended range is provided to allow further tailoring for customer needs.

6.0 AC OVERWEIGHT CHECK AMOUNT

The AC Overweight Check Amount collection parameter may be utilized by customers which use smaller volume AC bags than saline bags (e.g.: 500ml AC vs 1000ml Saline) and desire the AmiCORE to be configured to detect accidental bag switches. The default value for AC Overweight Check Amount is 0g which disables the feature to detect accidental bag switches. For customers wishing to enable this feature, perform the following steps to set this value:

- 6.1 Obtain the average weight of the completely filled AC container (bag/bottle) utilized by the customer. If a scale is unavailable, the AmiCORE scale may be used in Service Mode to obtain AC container measurements.
- 6.2 Multiply the average weight by 1.1 to add a 10% margin to the average weight.
- 6.3 Subtract the volume of the AC bag from the total from step 6.2. Enter the value into the AC Overweight Check Amount field.

Example:

Average AC Bag Weight	537 g
Average AC Bag Weight + 10%	591 g
AC Bag Volume	500 mL
AC Overweight Check Amount	91 g

The system will alert if the operator hangs a bag on the front scale during kit installation that weighs greater than the AC Volume plus the AC Overweight Check Amount (e.g.: 500ml AC Volume + AC Overweight Check Amount 91g = 591g). Note that the administrator may set up to three allowable preset AC volumes in Admin Mode but all volumes use the same AC Overweight Check Amount.

7.0 EASE OF SPOOL REMOVAL

After the system completes reinfusion, the separation pack is full of air and fluid which need to be partially emptied to allow for the spool to be more easily removed from the spool holder. The air and fluid are transferred from the separation pack to the IP bag via the Centrifuge Pump. By default, the system removes all the air from the pack but does not transfer any fluid. There are two service settings, **Pack Unload Volume** and **Pack Removal Rate** that control the volume and rate of the transfer stage; the following combinations of parameters are recommended:

	Pack Unload Volume (mL)	Pack Removal Rate (mL/min)	Result
Option 1 (default)	25	120	Air is removed from the separation pack but the fluid remains; minimal or no volume in the IP bag
Option 2	70	60	All air and fluid is moved out of the separation pack into the IP bag; the separation pack is empty
Option 3	35	120	Some fluid remains in the separation pack; some fluid is moved into the IP bag

8.0 IMPLEMENTATION

This technical service bulletin is applicable **ONLY** to AmiCORE Apheresis Systems located within Korea that are approved for upgrade to version 2.1M.

9.0 REVISION HISTORY

TSB Rev	Date	Change #	Name	Description of change
A	1 Nov 2024	CO-051576	J. Mashke	Initial TSB release.